# Adjoint and shape-gradient validation in JAX-DSMC

Every gate, study and finding, 2026-09-05 to 2026-10-11

2026-10-11

# How to read this deck

**Arbiters**, from strongest to weakest:

- analytic or exact references (closed forms, Rankine–Hugoniot, exact quadrature, replica-exact value gradients on collisionless cases);
- **CRN FD**: central finite difference on the *same* seeds (common random numbers), paired with the estimator seed by seed; noise grows like $1/h$ across hit-set jumps;
- **FD-only**: finite differences over many independent seeds (no pairing).

$z$ = (estimator − arbiter) / SE of the paired difference. "Seeds" always means independent forward runs. Lesson learned repeatedly: never conclude from 2 seeds.

**Timeline**

| dates | phase | part |
|---|---|---|
| 09-05 – 09-11 | forward gates; trajectory (DiCE) adjoint, boundary adjoints | I, II |
| 09-14 – 09-17 | estimator benchmark; P3 ensemble adjoint, route (b), virtual partners | II |
| 09-18 – 09-24 | embedded bodies; shape adjoint; level set; diffuse terms; cluster studies | III |
| 09-25 – 10-02 | `mode2` and squash studies on the cluster | III |
| 10-02 – 10-04 | bias study: seven findings, collisional terms built | IV |

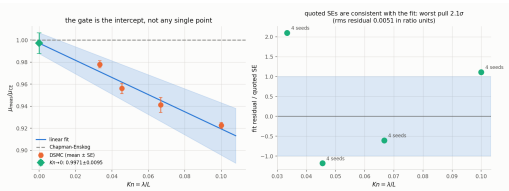
Derivation of the current estimator: `shape_derivation.tex` (with code: `shape_derivation_code.tex`).

# Forward gates: transport and shocks

**Shear viscosity, Couette** (Chapman–Enskog, no free parameter). Four wrong explanations of an apparent deficit preceded the right one; every one was a quantity read without an error bar. The gate is the double extrapolation, 4 seeds per point:

$$\mathrm{Kn} \to 0: \ 0.9971 \pm 0.0095, \qquad dx \to 0: \ +0.0135 \pm 0.0039, \qquad \mu/\mu_{\mathrm{CE}} = 1.0105 \pm 0.0103 \ (|t| = 1.0).$$

**2-D Taylor–Green decay**: viscosity $+0.38\,\%$ from Chapman–Enskog ($R^2 = 0.9994$).
**Normal shock** (Bird §12.11): $L/\lambda_1 = 5.98/6.12/6.18$ at $D = 1/2/3$ against $6.2$; jump conditions $0.1$–$0.7\,\%$, shock speed $0.3\,\%$.
**Inflow-driven Mach-2 shock**: all six moments within $0.5\,\%$ of Rankine–Hugoniot, mass flux to $0.06\,\%$.



# Forward gates: embedded bodies against SPARTA (2026-09-18)

Cut-cell geometry identical to SPARTA's (cell classes, cut set, volumes to $10^{-7}$); argon circle and sphere, 4 seeds each, every tracking diagnostic zero.

| gate | circle | sphere | criterion |
|---|---|---|---|
| final particle count | $z = -1.05$ | $z = 0.97$ | $\lvert z\rvert \le 2$ |
| total drag | $+0.31\,\%$ ($z = 1.36$) | $-0.11\,\%$ ($z = -0.30$) | $\lvert z\rvert \le 2$ |
| per-cell $n$, $u$, $T$ | rms $z$ 1.2, 90–92 % within 2 | rms $z$ 1.24, 91 % | |
| polygon vs analytic body | drag $+0.5\,\%$ | $+0.75\,\%$ | |

**Free-molecular drag** (FG5), extrapolated to zero blockage, against the half-range surface integral:

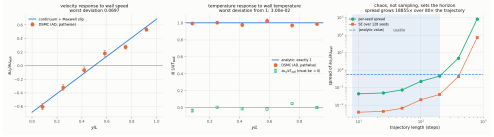
| case | measured $C_D$ | reference | $z$ |
|---|---|---|---|
| circle, specular | $17.080 \pm 0.080$ | 17.175 | $-1.18$ |
| circle, diffuse | $20.429 \pm 0.087$ | 20.429 | 0.00 |
| sphere, specular | $14.684 \pm 0.211$ | 14.804 | $-0.57$ |
| sphere, diffuse | $20.990 \pm 0.295$ | 21.211 | $-0.75$ |

The reference box (60 % blockage) raises the drag $\sim$50 % above the isolated body's.

# The trajectory (DiCE) adjoint, M6–M8 (2026-09-05 . . . 09-11)

Score-function plus pathwise AD through the whole run (`jax_dsmc/adjoint.py`).

- Oracles: [P1] Algorithm 2 and [P2] Algorithm 3 against `jax.grad` exactly ($10^{-11}$, $10^{-10}$); move→collide recursion to $10^{-16}$ (Oracle 3).
- Pathwise wall gradients: $\mathrm{d}J/\mathrm{d}T_{\mathrm{wall}}$, $\mathrm{d}J/\mathrm{d}u_{\mathrm{wall}}$ to $3 \times 10^{-12}$ against fixed-key FD, and against the Couette closed form (figure).
- Inflow `nrho` (score): FD/AD $= 1.04$, $|z| = 0.81$.
- **Attribution bias**: pairing each score with its own particle's reward is exact without collisions ($|z| = 0.18$) and $2.8\times$ low with them ($|z| = 22$), growing with the horizon (FD/AD $1.13 \to 2.82$ from 2 to 40 steps). Accommodation biased $2.8\times$; shock $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}u_1$ **wrong sign**.
- VSS $\alpha$: argon's $\alpha = 1.40$ sits at the stationary point $\sqrt{2}$ of $\langle \cos^2 \chi \rangle$; $\mathrm{d}T_y/\mathrm{d}\alpha \approx 0$ is the right answer (AD tracks FD, $|z| \leq 0.96$, across $\alpha$).

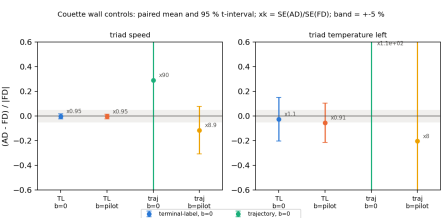


# Score estimators against CRN FD (review §4.8, 384 seeds)

Couette ($N = 256$, 10 steps) and inlet shock ($N = 128$, 4 steps):

| control | arm | AD − FD | $z$ | SE(AD)/SE(FD) |
|---|---|---|---|---|
| Couette $u_{\mathrm{wall}}$ | terminal-label | $-0.001 \pm 0.011$ | $-0.2$ | 0.95 |
| | trajectory, pilot baseline | $-0.068 \pm 0.11$ | $-1.2$ | 8.9 |
| | trajectory, no baseline | $+0.17 \pm 1.2$ | 0.3 | 90 |
| Couette $T_{\mathrm{left}}$ | terminal-label | $-0.004 \pm 0.03$ | $-0.3$ | 1.1 |
| | trajectory, no baseline | $+1.19 \pm 2.4$ | 1.0 | 106 |
| inlet $T$ | terminal-label | $+0.8\,\%$ | 2.6 | 1.0 |
| inlet $n_\rho$ | every score arm | — | — | **185–1300** |

- The trajectory score is unbiased where resolvable but its SE grows like $M^{2.8}$ in the horizon and $N^{0.56}$ in the particle count (it scores every event).
- A stochastically rounded count carries a score $\mp 1/f$: the reservoir density is out of reach of every score route. This decided the move to the ensemble adjoint (2026-09-15).



# P3 replication and the two quadratures of the ensemble adjoint

**P3 Test 4 (Couette walls)**, $N = 32\,000$, 32 seeds: $\mathrm{d}J/\mathrm{d}u_{\mathrm{wall}} = 3.032 \pm 0.010$ against the paper's $3.028$ — a replication at the paper's precision.
**P3 Test 3 (two-sided inflow)**: the transcription's own gradient is known to 1 % (1600-seed FD $0.1439 \pm 0.0014$); P3's retained-subset quadrature (a) depends on the retained count and on the reconstruction (affine $+18\,\%$, a Jensen bias of an affine fit to a convex field); the paper's $0.1517$ is **3–5 %** above every estimate of the transcribed model: a forward-model difference at the inlet, not an adjoint one.
**Route (b): values carried on the particles**, realised pairs as the collision quadrature (one interpolated ghost per collision). On P3's model, two steps with strong collisions: interpolation-free (b) against CRN FD $+0.0008 \pm 0.0015$ (**0.5 %** resolution); route (a) with every particle retained $-0.8\,\%$ ($3\sigma$). $\mathrm{N}_0 = 16\,000$: (b) $0.1469 \pm 0.0005$ vs FD $0.1439 \pm 0.0014$ ($z = 2.0$) / $0.1466$ vs $0.1453$ ($z = 0.8$). Cost $1/40$ of (a).

**On the production solver** (VSS, NTC, move-then-collide, stochastic counts), inlet $T$, $N = 128$: the affine ghost basis read $-8.3\,\%$ ($z = -4.3$; collision term 27 % of the gradient); adding the $|v|^2$, $x^2$ columns (quadratic basis) gives $z = -0.95$. Reservoir density: **0.43× the FD's noise**, unbiased.

# Walls as kernels, kernel parameters, and the exact sequential pull-back

**Diffuse walls**: the realised re-emission is a sample of the kernel, so wall $T$, $u$ and accommodation enter only through the score of the re-emission density; no $\varepsilon$. **Kernel parameters** (`diam`, `omega`): $\partial_\theta \log q$ on the accepted pairs times their four-term.

**The $\Delta t$-linearised pull-back is biased** where particles collide more than once a step: relaxation `diameter` $-9.9\,\%$ ($z = -4.2$) at 0.16 collisions per particle-step. **The sequential pull-back** (accepted attempts in reverse slot order, ghosts from the field as it stood after each) is the exact discrete adjoint of NTC: $-0.667 \pm 0.039$ vs FD $-0.675 \pm 0.017$.

Every control, sequential pull-back (review §5.7), CRN FD $h = 0.1$:

| control | ensemble | FD | $z$ | SE ratio |
|---|---|---|---|---|
| Couette accommodation | $0.2917 \pm 0.009$ | $0.2615 \pm 0.017$ | 1.7 | 0.52 |
| Couette $u_{\text{wall}}$ ($\varepsilon = 0$) | $0.5865 \pm 0.014$ | $0.5933 \pm 0.0093$ | $-0.53$ | 1.47 |
| Couette $T_{\text{left}}$ | $0.1634 \pm 0.0088$ | $0.1804 \pm 0.012$ | $-1.2$ | 0.73 |
| inlet $n_\rho$, $N = 128$ | $1.375\text{e}5 \pm 1.3\text{e}3$ | $1.377\text{e}5 \pm 2.5\text{e}3$ | $-0.08$ | 0.53 |
| inlet $T$, $N = 256$ | $1.364\text{e}5 \pm 3.6\text{e}3$ | $1.374\text{e}5 \pm 1.1\text{e}3$ | $-0.32$ | 3.3 |
| relaxation `diameter` | $-0.663 \pm 0.026$ | $-0.6723 \pm 0.012$ | 0.32 | 2.1 |
| relaxation `omega` | $0.2824 \pm 0.010$ | $0.2607 \pm 0.014$ | 1.2 | 0.69 |

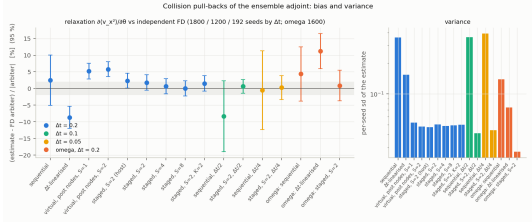
The accommodation FD is a step function of a decision parameter and cannot arbitrate below $\sim 3\,\%$.

# Virtual-partner pull-back: bias and variance (review §5.12)

The sequential form's noise grows with the collisions per particle-step and blows up above $\sim 0.5$. P3's virtual-partner average, with NTC weights, all four values from one cell fit, $S$ substeps. Relaxation `diameter` against a 1800-seed FD-only arbiter $-0.6800 \pm 0.0034$:

| pull-back | estimate | per-seed sd | bias |
|---|---|---|---|
| sequential | $-0.663 \pm 0.026$ | 0.36 | $+2.5\,\%$ $(0.7\sigma)$ |
| $\Delta t$-linearised | $-0.739 \pm 0.011$ | 0.15 | $-8.6\,\%$ $(5\sigma)$ |
| virtual, end-of-step nodes, $S = 2$ | $-0.641 \pm 0.007$ | 0.048 | $+5.8\,\%$ $(5\sigma)$ |
| virtual, staged nodes, $S = 2$ | $-0.668 \pm 0.007$ | 0.050 | $+1.8\,\%$ |
| virtual, staged, $S = 4$ | $-0.675 \pm 0.007$ | 0.048 | $+0.7\,\%$ |
| virtual, staged, $S = 8$ | $-0.680 \pm 0.007$ | 0.049 | $+0.1\,\%$ |

- Residual $O(\Delta t/S)$; per-seed sd $7\times$ below sequential at 0.16 coll./step, 37–56$\times$ at 0.8.
- Staged nodes (the slot-order state at each substep) remove the frozen-field bias.
- Shipped: $(S, K) = (4, 8)$, staged.

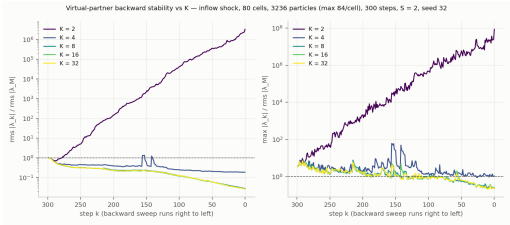


# Partner count, stability, and the Rankine–Hugoniot acceptance case

**Stability** (300-step shock, $\leq 84$ particles per cell): $K = 2$ grows geometrically ($3 \times 10^6$ over the horizon); $K = 4$ marginal; $K \geq 8$ one decaying curve. The backward is refused when rms $\lambda$ grows more than $100\times$.

**Acceptance case** $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}M_1$, 300 steps, where the pathwise derivative had the wrong sign: ensemble $2.74 \pm 0.66$, CRN FD $2.54 \pm 0.17$ / $2.17 \pm 0.42$, $z = 0.3$. Right sign; coarse (per-seed sd $4\times$ the FD's): long transients cost variance; prefer windowed, quasi-steady objectives.

Virtual pull-back, every control (staged, $S = 2$, $K = 8$): inlet $T/n_\rho$ at 0.8 collisions per particle-step $z = -0.7/-0.4$, now below the CRN FD's own noise; Couette $u_{\mathrm{wall}}$ $z = +0.8$, $T_{\mathrm{left}}$ $+1.7$; `omega` $+0.9\,\%$ against a 1600-seed arbiter.



# First shape oracles (2026-09-17/18)

**Planar oracle** (one step, collisionless, specular wall): analytic $\mathrm{d}J/\mathrm{dgap} = -0.5917$. The pathwise gradient at frozen hits reads **0** (the hit set is invisible to it); the value-weighted $\varepsilon$-hit score $0.588 \pm 0.008$.
**G3a window score** (1-D closed box, windowed impulse): the $\varepsilon$ hit-timing score has no resolvable mean on a steady window and costs 4–8$\times$ the FD's sd $\Rightarrow$ the shape runs use $\varepsilon = 0$; the hit set enters through the replayed wall map and band terms instead.

**Value-form rewards on Couette** (48 seeds, CRN FD):

| gate | objective, control | ensemble vs FD | $z$ |
|---|---|---|---|
| G1 | windowed $v_y^2$, wall speed | $+0.491 \pm 0.025$ vs $+0.4925 \pm 0.016$ | $-0.05$ |
| G2 | windowed shear (per-hit), wall speed | $-0.0177 \pm 0.0014$ vs $-0.0165 \pm 0.0008$ | $-0.77$ |
| G2 | same, $T_{\mathrm{left}}$ | $-0.0025 \pm 0.0006$ vs $-0.0032 \pm 0.0015$ | $+0.46$ |

The first G2 read $40\times$ too small: the weight of a hit's kernel score must include the hit's own reward (reward-to-go from the draw onward).

# G3: a translated wall in a 2-D Couette, and what it taught

1. **Hits change step.** Windowed $v_y^2$: pathwise alone $+0.82 \pm 0.23$ vs FD $+0.05 \pm 0.15$; the hit-timing band term ($-0.43 \pm 0.02$) closes it.
2. **Two-sided fits are biased next to a wall.** Windowed shear came out with the wrong sign ($-0.012$ vs $+0.017$); nodes restricted to the leaving side ($v \cdot n > 0$) fix the sign.
3. **Bias or horizon?** With replica-exact value gradients the estimator equals the *truncated* FD on both objectives ($z = 0.5$); the gap to the full FD is the captured horizon (72 captured steps before the window close it).
4. **Estimators of $\nabla_x \lambda$** against replica-exact truth (windowed $v_y^2$, exact $x/v$ part $+0.374 \pm 0.111$): one-sided $-0.29 \pm 0.17$ bias, pooled $-0.20 \pm 0.11$, one-sided + along-flight $-0.08 \pm 0.11$, pooled + along-flight $-0.07 \pm 0.11$. The along-flight identity holds per particle ($-1.3\text{e}{-5} \pm 2.8\text{e}{-5}$ on a scale $1.6\text{e}{-3}$).
5. **Unresolved**: the $x/v$ part of a *translated flat wall* (a window-end delay sensitivity); every fitted or value-based form reads 0.

**Decisions (2026-09-20)**: shape controls are normal displacements of curved surfaces; the body is a $C^1$ level set (a polygon's vertex term is a representation artefact).

# Circle radius, level set and normal modes (2026-09-18 . . . 09-21)

**Circle radius**, analytic body, 48 seeds, collisionless:

| term | $10^{-20}$ N/m |
|---|---|
| tally | $1.28 \pm 0.01$ |
| $\nabla_v \lambda$ part | $0.02 \pm 0.02$ |
| $\nabla_x \lambda$ part | $0.14 \pm 0.09$ |
| **ensemble** | $1.44 \pm 0.09$ |
| CRN FD | $1.395 \pm 0.02$ |
| analytic $F/r$ (no blockage) | $1.23$ |

$z = +0.5$. The tally reproduces $F/r$ to 4 %; the rest is the blockage feedback.

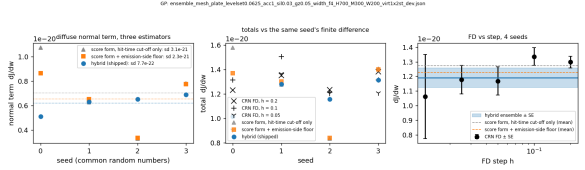
**Level set**: the same simulation as the analytic body to the surface error (objective to $2 \times 10^{-6}$); radius $z = -1.4$ (4 seeds).
$\cos 2\theta$ **mode** (coarse, 4 seeds): ensemble $-1.587 \pm 0.030$ vs FD $-1.617/-1.571$ ($z = +0.7/-0.2$); per-seed sd 3.8 % (radius: 45 %).
**Sphere radius** (8 seeds, cluster): $3.389 \pm 0.048$ vs FD $3.487 \pm 0.031$ ($10^{-20}$), $z = -1.71$.
**Lane vs stencil tracker**: gradients agree to $2.3 \times 10^{-15}$.
**First optimisation**: 8 GD iterates, drag $-7.4\,\%$ at fixed area.

# Grazing hits and the diffuse terms (2026-09-21/22)

**Grazing regularisation**: the per-hit derivative carries $1/(v \cdot n)$; one hit at $|v \cdot n|/|v| = 3 \times 10^{-3}$ was **15 %** of a window's tally. Floor in the derivative only:

| $\epsilon$ | 0 | 0.01 | 0.03 | 0.1 | 0.3 |
|---|---|---|---|---|---|
| seed sd of the tally term | 8.4 % | 5.9 % | 4.2 % | 2.9 % | 2.1 % |
| paired bias vs $\epsilon = 0$ | — | $+1.0 \pm 1.1$ | $+0.5 \pm 1.8$ | $-1.2 \pm 2.3$ | $-7.4 \pm 2.4$ |

**Diffuse capsule plate (GP)**, control = half-width: first pass 80 % of FD (silhouette term needed; score resolution); then the **hybrid normal term** (pathwise reward through the reparameterised re-emission + score on the value after the hit) cut the normal term's seed sd $4\times$: $1.191 \pm 0.068$ **vs FD** $1.169 \pm 0.096$ ($z = +0.19$; $|z| \leq 1.6$ at five FD steps). Silhouette = **45 %** of the gradient.
**Diffuse circle (GD)**: $1.348 \pm 0.25$ with timing vs FD $1.442 \pm 0.03$ / $1.128 \pm 0.16$; silhouette 61 %; 9 % low against the precise FD without the timing term (open then; see Part IV).



# The specular capsule and the future-class fit (2026-09-22, superseded)

**Specular capsule**: mixed-pool fits read $-27\,\%$ ($z = -11$); the tally alone matched the FD seed by seed, and the deficit sat in the $\nabla\lambda$ parts. Cause: at the curved cap the one-sided rule does not separate leaving particles from those heading for the flat face.
**Stratifying the fits by "a body hit at a later step"** fixed it: $0.964 \pm 0.037 \to 1.356 \pm 0.011$ vs FD $1.35$ ($z = +0.08$), and became the default.

**It was biased by construction** (found 2026-09-25, explained 2026-10-02): the class is a property of the particle's future, so the fit drops $\nabla P\,(\lambda_1 - \lambda_0)$ (Part IV, Finding 1). On the capsule $P \approx 0$ after a reflection, so it was harmless there; on the fine $\cos 2\theta$ mode (32 seeds, cluster) it read $-2.386 \pm 0.005$ against FD $-1.592$ ($z \approx -120$), the tally alone matching the FD ($z = -0.33$).

**Consequences**: the `optc` optimisations ran with the class; the squash studies did not. The class is now a diagnostic option that warns.

# Cluster studies: optc free-shape optimisation (2026-09-22 . . . 09-25)

Coarse circle, $f_{\mathrm{num}} \times 1$, specular, collisionless, fixed area; $S \in \{1, 2, 4, 8\}$ CRN seeds per iterate $\times$ {GD, Adam}; along-pool fits *with the future class*.

| run | iterates | drag change (in-sample) | held-out (4 seeds) | end |
|---|---|---|---|---|
| S1 GD | 131 | $-14.9\,\%$ | within 1 % | cancelled |
| S2 GD | 216 | $-19.2\,\%$ | within 1 % | time limit |
| S4 GD | 113 | $-19.0\,\%$ | within 1 % | time limit |
| S8 GD | 61 | $-13.9\,\%$ | within 1 % | host OOM |
| S1 Adam | 250 | $-37.1\,\%$ | within 1 % | complete |
| S2 / S4 / S8 Adam | 96 / 102 / 37 | $-38.8 / -35.0 / -41.8\,\%$ | within 1 % | inflow face / limit |

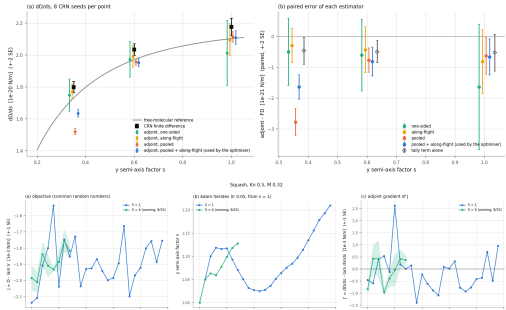
- Every failure on the way was a real tracking or meshing defect, fixed at its root (reinitialisation band, smoothed Adam step, stuck particles, concave re-entry, split cells, replay tolerance $10^{-9}$), and led to the lane tracker ($2\times$ faster, $-45\,\%$ memory).
- The Adam bodies reach the outflow and inflow faces: their reductions are not those of an isolated body (a clearance constraint is needed). The gradient they followed was class-biased.

# Cluster studies: the squash gate and optimisations (2026-10-01/02)

**Free-molecular gradient gate** ($f_{\mathrm{num}} \times 1$, 8 seeds), adjoint − CRN FD (% of FD):

| variant | $s = 0.35$ | 0.6 | 1 |
|---|---|---|---|
| tally alone | $-2.6$ | $-2.4$ | $-2.4$ |
| `one` | $-2.8$ | $-3.0$ | $-7.5$ |
| `one_pool` | $\mathbf{-15.5}$ | $-3.8$ | $-2.9$ |
| `along` | $-1.6$ | $-2.1$ | $-3.8$ |
| `along_pool` (used) | $\mathbf{-9.1}$ | $-4.0$ | $-3.0$ |

($\pm 1$–$5\,\%$; `along_pool` $z = -8.3$ at $s = 0.35$.)
**Optimisations** (Adam, 25 iterates, `along_pool`, 0.2-step timing band, no cut-volume or order terms, identity without $\mathcal{L}^*\lambda$): free-molecular S1/S4: $s = 0.355/0.339$ (reference optimum 0.35); Kn 0.5: S1 drifts to $s = 1.21$, S4 to $1.13$ (8 iterates); Kn 0.5, M 10: S1 to $0.48$ (12 iterates).



# Bias study (2026-10-02 . . . 10-04): method

Small runs on the 4 GB workstation GPU; every estimator paired with the CRN FD of the same seed (`scratch/bias/`: `kn_gate.py`, `fm_exact.py`, `horizon.py`, `fd_only.py`, `summarise.py`).

- Squash ellipse ($R = 1.5$, $h_\phi = 0.0625$, grazing 0.05), 20×20 cells, warm-up 700, 300 captured steps, window 200, CRN FD $h = 0.05$.
- Free-molecular and Kn $= 0.5$ ($\nu \Delta t \approx 2.5\,\%$) at $f_{\mathrm{num}} \times 2$ (75 particles per cell); virtual pull-back $2 \times 4$ locally.
- All $\nabla_x \lambda$ variants from one replay, so their differences are paired too.

**An exact free-molecular reference that failed**: without collisions a path is a deterministic function of $\theta$, so the forward-mode JVP chained step to step is the exact pathwise derivative (reproduces the drag to $2 \times 10^{-16}$). The box is a dispersing billiard (convex body, specular walls): tangents grow at every bounce, and the chained JVP ranged from $-5.5\times$ to $+8.8\times$ the tally term over 8 seeds. Useless per seed; the fits' smoothing is what makes the ensemble adjoint usable.

# Audit: every $\theta$-dependence of the forward

| dependence | term | before the study | now |
|---|---|---|---|
| hit times, points, specular $v^{\mathrm{out}}$ | pathwise $x$, $v$ | in | in |
| reward through $v^{\mathrm{out}}$ | tally | in | in |
| diffuse kernel normal | score | in | in (0 specular) |
| hit $\leftrightarrow$ miss | silhouette | in | in (off for specular) |
| hit crosses a window edge | timing | **reported, not added** | in, 10-step band (F6) |
| hit/collision order at every step | order | **missing** | in (F7) |
| cut-cell volume in NTC | operator | **missing** | in (F2) |
| $\nabla_x \lambda$ along $v$ with collisions | along-flight | $\mathcal{L}^* \lambda$ **dropped** | in (F3) |
| node sets of the fits | — | **future class (optc); pool across body** | state-based (F1, F5) |
| newborn / leaving particles' hits | — | unmonitored | warns above $10^{-3}$ (F4) |

The VJP of the move is complete: every floating output of the wall map gets a cotangent, and the replay is checked against the record at every step. F1–F7: the findings that follow.

# Finding 1: the future-class fit is biased, and cannot be repaired

Coarse circle, level-set body, `mode2` ($\cos 2\theta$), 4 CRN seeds (32–35), the same records for every row; FD $= -1.594 \pm 0.043$, tally term $-1.596$ ($\times 10^{-20}$), so the $\nabla\lambda$ parts should sum to FD−tally $= +0.01 \pm 0.68 \times 10^{-21}$.

| fit class | $x$ part $10^{-21}$ | $v$ part $10^{-21}$ | gradient $10^{-20}$ | gradient − FD (paired) $10^{-20}$ |
|---|---|---|---|---|
| no class (`none`) | $-0.28$ | $+0.36 \pm 0.15$ | $-1.587 \pm 0.030$ | $+0.007 \pm 0.071$ |
| hit-ahead, current | $-0.72$ | $-6.02 \pm 0.34$ | $-2.269 \pm 0.036$ | $-0.675 \pm 0.057$ |
| hit-ahead, stopped at the next birth | $-0.80$ | $-10.82 \pm 0.62$ | $-2.758 \pm 0.069$ | $-1.164 \pm 0.091$ |

**Refilling is not the cause.** A purer class makes the bias *larger*. The stratified fit estimates $\nabla\lambda_\kappa$ within each class, while

$$\lambda = P\lambda_1 + (1-P)\lambda_0 \;\Rightarrow\; \nabla\lambda = P\nabla\lambda_1 + (1-P)\nabla\lambda_0 + \underbrace{\nabla P\,(\lambda_1 - \lambda_0)}_{\text{dropped}},$$

with $P(z)$ = probability of a later body hit. A reflected particle can come back, most likely after a specular $y$-wall bounce (not checked), so $\nabla_v P \neq 0$, and $\lambda_1 - \lambda_0 \approx$ the next hit's drag. Neither the silhouette nor the timing term carries this. On the capsule, $P \approx 0$ after a reflection, so the class was harmless there.

**Not repairable**: restoring $\nabla P(\lambda_1 - \lambda_0)$ needs $\nabla P$ from the same local regression, and that sum is the mixed-pool fit again, with more variance. Free-molecular squash: $+4$ to $+20\,\%$. Valid stratification uses the present state ($v \cdot n$, the tangent half-space).

# Free-molecular specular squash: adjoint − FD, % of FD

| variant | $s = 0.35$ | | $s = 0.6$ | | $s = 1$ | |
|---|---|---|---|---|---|---|
| | no timing | + timing | no timing | + timing | no timing | + timing |
| `one` | $-3.0 \pm 0.8$ | $-2.1 \pm 1.2$ | $+1.3 \pm 4.1$ | $+1.3 \pm 3.9$ | $-1.8 \pm 2.3$ | $-1.7 \pm 2.3$ |
| `one_pool` | $-19.4 \pm 0.7$ | $-18.4 \pm 1.1$ | $-5.1 \pm 1.3$ | $-5.1 \pm 1.8$ | $-2.3 \pm 1.5$ | $-2.3 \pm 1.6$ |
| `along` | $-1.7 \pm 1.2$ | $-0.7 \pm 1.7$ | $-0.7 \pm 2.4$ | $-0.7 \pm 2.4$ | $-0.9 \pm 1.5$ | $-0.9 \pm 1.5$ |
| `along_pool` | $-9.9 \pm 1.3$ | $-9.0 \pm 1.7$ | $-4.5 \pm 1.4$ | $-4.5 \pm 1.9$ | $-2.1 \pm 1.4$ | $-2.1 \pm 1.6$ |
| `one_pool_plane` | $-2.7 \pm 0.6$ | $-1.7 \pm 0.9$ | $-2.6 \pm 1.2$ | $-2.6 \pm 1.9$ | $-1.5 \pm 1.5$ | $-1.5 \pm 1.6$ |
| `along_pool_plane` | $-1.7 \pm 1.1$ | $-0.7 \pm 1.5$ | $-2.5 \pm 1.4$ | $-2.5 \pm 1.9$ | $-1.5 \pm 1.4$ | $-1.4 \pm 1.5$ |

8 CRN seeds per $s$ ($f_{\mathrm{num}} \times 2$), $\pm 1$ SE of the paired difference. "+ timing": window-edge term (0.2-step band at $s = 0.35, 0.6$; 10-step band at $s = 1$).

- `one_pool`, `along_pool` (the cluster runs' variant): $-19\,\%$ / $-9\,\%$ at $s = 0.35$, $-5\,\%$ at $s = 0.6$ — Finding 5.
- Restricted to the hit's half-space (`*_plane`): every entry within $2\sigma$; pooled over $s$ a residual of about $-2\,\%$ remains (located below).

# Finding 5: the pooled fit reaches across a thin body

The pooled fit draws its $k = 96$ nodes from the $3 \times 3$ block around the hit's cell ($1.5 \times 1.5$ m). At $s = 0.35$ the body is $2sR = 1.05$ m thick, two cells: the block around a hit on the upper face holds the particles *below* the body. Their values have nothing to do with the query's, and the one-sided rule does not separate them (below the body, $v \cdot n_{\text{top}} > 0$ means moving towards the lower face).

**Fix**: drop the nodes more than a quarter cell behind the plane through the query, normal to its hit normal, $(x_{\text{node}} - x_q) \cdot n \geq -\frac{1}{4}\Delta x$. Variants `one_pool_plane`, `along_pool_plane`; the latter is the default everywhere. It also removes most of the $v$ part's spread ($-1.1 \to -0.12 \times 10^{-21}$ at $s = 0.35$).

# Finding 6: the window-edge timing term

- It was left out as "zero mean on a steady window". The window is steady (drag change over the 200 steps $-0.1\,\%$ specular, $+0.8\,\%$ diffuse).
- The 0.2-step band (one step per edge, ∼10 hits per seed) read $+2.8 \pm 1.2\,\%$ at $s = 1$ and $-35\,\%$ on one diffuse seed: first mistaken for the missing $-2.4\,\%$; it is noise.
- 10-step band (whole steps per edge), same 8 seeds: $+0.03 \pm 0.25\,\%$.

**Decision**: the term belongs to the windowed objective's derivative (nonzero whenever the window is not steady), so it is in the reported gradient, with the 10-step band as default.

**Diffuse squash** ($a = 1$, free-molecular, $s = 1$, 8 seeds, silhouette band 0.03):

| variant | $s = 1$ | |
|---|---|---|
| | no timing | + timing |
| `one` | $+1.0 \pm 3.1$ | $+1.2 \pm 3.0$ |
| `one_pool` | $-6.0 \pm 3.1$ | $-5.8 \pm 3.0$ |
| `along` | $-0.6 \pm 3.4$ | $-0.4 \pm 3.3$ |
| `along_pool` | $-3.2 \pm 3.4$ | $-3.1 \pm 3.3$ |
| `one_pool_plane` | $-3.1 \pm 3.1$ | $-2.9 \pm 3.0$ |
| `along_pool_plane` | $-1.6 \pm 3.4$ | $-1.4 \pm 3.3$ |

$9.12$ (tally, hybrid) $+ 9.84$ (silhouette) $- 0.13$ (normal score) $- 0.15$ ($\nabla\lambda$) $[10^{-21}]$ against FD $19.0$: the silhouette term is half the gradient; every variant but the cross-body `one_pool` within its $\pm 3\,\%$ noise. The first diffuse squash gate.

# Finding 2: the cut-cell volume term, built and gated

NTC's $1/V_c$ was seen by neither the replay nor the VJP. Built 2026-10-02 (derivation: Term 7).
**Machinery gate** (Kn $= 0.5$, $f_{\mathrm{num}} \times 4$, 8 seeds): scale the NTC volume of a set of cells by $1+\epsilon$ — the only change — and compare the FD in $\epsilon$ with the adjoint's $\sum_c g_c V_c$:

| cells scaled | FD | adjoint | $z$ |
|---|---|---|---|
| all | $2.11 \pm 0.26$ | $2.45 \pm 0.08$ | $-1.2$ |
| cut cells | $0.69 \pm 0.45$ | $0.34 \pm 0.01$ | $+0.8$ |

($10^{-3}$ N/m)

**Geometry factor**: $\sum_c \partial V_c/\partial s = -7.068$ against $-\pi R^2 = -7.069$.
**Size** at Kn 0.5, $s = 1$: $-4.4 \times 10^{-4} = -2.9\,\%$ of $\mathrm{d}D/\mathrm{d}s$ (seed SE 0.1 %).

# Finding 3: the along-flight identity with collisions

The identity used was $v \cdot \nabla_x \lambda = -T$; with collisions it is $-T - \mathcal{L}^* \lambda$ (the omission was justified at $\leq 0.3\,\%$ collisions per step; Kn 0.5 has 2.5 %). Fitted slope against the identity's value of its component along $v$, paired (x part, $10^{-4}$ N/m, 8 seeds):

$$\text{fit} - \text{identity}(T + \mathcal{L}^* \lambda) = -1.3 \pm 0.5, \qquad \text{fit} - \text{identity}(T) = +4.9 \pm 0.5.$$

- Without $\mathcal{L}^* \lambda$ (the cluster runs) the identity is off by 3.2 % of $\mathrm{d}D/\mathrm{d}s$ ($z \approx 10$); with it fit and identity nearly agree; the residual $-0.8\,\%$ is the $O(\nu \Delta t)$ of the discrete scheme.

Kn 0.5, adjoint − paired FD (% of FD; the $h = 0.05$ FD has 18 % per-seed noise):

| variant | $s = 1$ | |
|---|---|---|
| | no timing | + timing |
| `one` | $-7.3 \pm 6.1$ | $-7.2 \pm 6.2$ |
| `one_pool_plane` | $-8.0 \pm 6.0$ | $-7.8 \pm 6.1$ |
| `along` | $-6.2 \pm 5.9$ | $-6.0 \pm 6.0$ |
| `along_free` | $-9.9 \pm 5.9$ | $-9.8 \pm 6.0$ |
| `along_pool_plane` | $-7.2 \pm 6.1$ | $-7.1 \pm 6.2$ |
| `along_pool_plane_free` | $-11.0 \pm 6.1$ | $-10.8 \pm 6.1$ |

(`_free`: identity without $\mathcal{L}^* \lambda$.)

# Finding 7: the order of a hit and the step's collisions

NTC collides once per step, after the move. A hit that slips from step $k$ into $k+1$ collides in step $k$ as the incoming particle instead of the reflected one: $J_c = c(x_w, v^{\mathrm{in}}) - c(x_w, v^{\mathrm{out}})$, $c = \Delta t\, \mathcal{L}^* \lambda$, at every step boundary. The window-edge reward jump is the collisionless remainder of the same event.

Kn 0.5, $s = 1$, 8 seeds: $+6.3 \pm 4.4 \times 10^{-5} = +0.4 \pm 0.3\,\%$ of $\mathrm{d}D/\mathrm{d}s$. Small, because the incoming and reflected collision effects differ by a small fraction of the reward; in the gradient by default.

**Finding 4** (unscored hits): hits of particles born in the step or leaving the box in it get no reward in $\lambda$ and no VJP. Their share is recorded; zero while the body is more than $|v|\Delta t$ from the inflow and outflow faces; nonzero only for the `optc` Adam bodies that reached those faces. Now warns above $10^{-3}$.

# Audit: where could a no-collision assumption hide?

The step is move (hits) → NTC collisions. Each term's justification, re-checked with collisions:

| term | assumption | with collisions |
|---|---|---|
| pathwise $x$, $v$ | $\nabla\lambda_k$ at the post-move state; $\lambda_k$ includes step $k$'s collisions | holds |
| tally, score, hybrid normal | exact reward; weights $\lambda$ + later rewards | holds |
| silhouette | straight flight to the step end | holds (collisions follow the move) |
| **timing band** | a hit changing step matters only at the window edges | **wrong**: the hit/collision order swaps — Finding 7 |
| along-flight | $v \cdot \nabla_x \lambda = -T - \mathcal{L}^* \lambda$ | $O(\nu\Delta t)$ in the discrete scheme (F3) |
| cut-cell volume | — | built (F2) |
| cell membership of a hit particle | omitted (integer) | $O(\nu\Delta t)$ below the order term |
| fits at hit slots only | only hit slots' post-move states depend on $\theta$ | holds |

# The free-molecular $-2\,\%$: noise or bias?

Adjoint $-$ FD (%), 8 seeds per $s$, $f_{\mathrm{num}} \times 2$; pooled with inverse-variance weights:

| | $s = 0.35$ | $0.6$ | $1$ | pooled |
|---|---|---|---|---|
| `along_pool_plane` | $-1.7$ | $-2.5$ | $-1.5$ | $-1.8 \pm 0.7$ ($z = -2.5$) |
| `one_pool_plane` | $-2.7$ | $-2.6$ | $-1.5$ | $-2.6 \pm 0.5$ ($z = -5.3$) |
| tally term alone | $-1.2$ | $-2.6$ | $-1.2$ | |
| tally alone, $f_{\mathrm{num}} \times 1$ (cluster gate) | $-2.6$ | $-2.4$ | $-2.4$ | ($\pm 0.9$–$1.4$ each) |

**Systematic**, and it sits in the *tally* term, which is exact given the hits.

- FD curvature ($h = 0.05$, on the analytic reference): $\leq 0.15\,\%$, wrong sign.
- Grazing cut-off removed (same forward): $+0.15 \pm 0.25\,\%$.
- **The captured horizon**: particles reflected *before* the captured steps bounce off the specular $y$ walls ($\sim$230-step round trip) and re-hit inside the window; the adjoint cannot see their first hit's dependence on the shape. Capturing 600 instead of 300 steps (same forward, $f_{\mathrm{num}} \times 4$, 8 seeds): $+0.8 \pm 0.25\,\%$ at $s = 1$, $+1.1 \pm 0.3\,\%$ at $s = 0.35$.

Most of the $-2\,\%$ is truncation, not a code error; it grows with the box's memory (reflective walls, trapping bodies). Gates at the 1 % level: $\geq 600$ captured steps or open $y$ faces. Cluster jobs `fm_hor_s*` capture the whole run to close the remainder.

# Kn $= 0.5$: the remaining gap, and what it is not

**Precise FD** (FD-only, 32 seeds, $f_{\mathrm{num}} \times 2$): $h = 0.1$: $0.01650 \pm 0.00019$; $h = 0.2$: $0.01605 \pm 0.00010$; Richardson $\mathbf{0.01665 \pm 0.00026}$.
**Adjoint** (8 seeds, every term): `along` $-6.5\,\%$, `along_pool_plane` $-7.5\,\%$, `one_pool_plane` $-8.3\,\%$ (each $\pm 1.8$); the old configuration $-9.7\,\%$.

**Ruled out** (paired, same forward):

- horizon 300 $\rightarrow$ 450 captured steps: $+0.1 \pm 0.4\,\%$ (fit variants);
- virtual quadrature $S \times K = 2 \times 4 \rightarrow 4 \times 8$: $< 1\,\%$;
- fit width $k = 48, 96, 192$ pooled: within 1 % (unpooled $k = 16$: $+4$–$10\,\%$, noisy);
- one-sided fits in the backward next to the body: $< 1\,\%$; timing band: $+0.15\,\%$;
- the sequential (exact) pull-back as referee: $50\times$ noisier at this collision rate — useless.

**Open**: tally + volume alone $\approx$ the FD; the fitted $v$ part ($-10\,\%$ of $\mathrm{d}D/\mathrm{d}s$) is the prime suspect. Cluster `biasgate` jobs: $f_{\mathrm{num}} \times 1$, $4 \times 8$, 64-seed FD, whole-run horizon.

# What the week's changes do to the squashkn result

The squashkn cluster runs (2026-10-01/02) used the then-default estimator. Same 8 Kn 0.5 seeds, $s = 1$, paired, % of the old gradient:

| change | effect on $\mathrm{d}D/\mathrm{d}s$ | sign of the old error |
|---|---|---|
| $\mathcal{L}^*\lambda$ in the along-flight identity (F3) | $+4.1 \pm 0.2\,\%$ | too low |
| half-space restriction of the pooled fit (F5) | $+0.8 \pm 0.05\,\%$ | too low |
| cut-cell volume term (F2) | $-2.9 \pm 0.1\,\%$ | too high |
| hit/collision order term (F7) | $+0.4 \pm 0.3\,\%$ | too low |
| 10-step timing band (F6) | $\approx 0$ in the mean; far less noise | — |
| **net** | $+\mathbf{2.4}\,\%$ ($+2.0 \pm 0.16$ without F7) | |

Against the precise FD: old $-9.7\,\%$ $\rightarrow$ now $-7.5\,\%$ (the open gap).

**On the optimum**: $J'(s) = \mathrm{d}D/\mathrm{d}s - \lambda_A\,\mathrm{d}A/\mathrm{d}s$ is the difference of two nearly equal numbers ($\lambda_A\,\mathrm{d}A/\mathrm{d}s = 1.537 \times 10^{-2}$, $\mathrm{d}D/\mathrm{d}s \approx 1.47 \times 10^{-2}$ at $s = 1$), so a few per cent of $\mathrm{d}D/\mathrm{d}s$ moves the zero of $J'$ a long way. A linear fit of the old runs' $\mathrm{d}D/\mathrm{d}s$ in $s$ ($\pm 50\,\%$ on the slope) puts the old zero at $s^* \approx 1.08$–$1.11$, the corrected one at $\approx 1.04$–$1.05$, and one matching the FD ($+8\,\%$) at $\approx 0.92$–$0.94$. The Kn 0.5 optimum is ill-conditioned; re-run squashkn only after the gap is closed. M 10: the corrections were not measured there.

# Why the interpolant was reopened (2026-10-09/10)

The virtual pull-back asks $(3+2K)N$ field values per substep ($19N$ at $4\times 8$): the node's own value, the two post-collision ghosts of each virtual partner, the boundary move. Until 2026-10-11 each was a **per-query** weighted quadratic fit over the $k$ nearest nodes of the query's cell.

**Its answer depended on** $k$ (Kn 0.5 squash, $s=1$, $f_{\mathrm{num}}\times 4$, paired, % of $\mathrm{d}D/\mathrm{d}s$):

| $k$ | 16 | 32 | 48 | 64 | 96, 128 |
|---|---|---|---|---|---|
| vs $k=128$ (3 seeds) | diverges | $-3.2\pm 0.3$ | $0.0\pm 0.7$ | $+0.1\pm 0.1$ | 0 (whole cell) |
| seconds per seed | 465 | 670 | 790 | 985 | 1295, 1660 |

Over 12 seeds: $k=32$ $-3.5\pm 1.8\,\%$ vs FD, $k=64$ $-2.0\pm 2.0\,\%$ ($k$ 32→64: $+1.6\pm 0.7\,\%$, half the 3-seed estimate). Cost: $\approx 300$ s fixed $+\ 10.7$ s per unit of $k$; at $k\geq$ occupancy the fit is the whole cell and its weights $1/(1+(d/d_k)^2)$ only span $[\frac{1}{2},1]$.

**Idea**: if the fit spans the cell anyway, fit *once per cell* ($C$ fits, not $19N$) and evaluate the polynomial at every query. Settled first: $S$, $K$ (S1/S2/S8, K4/K16 within $\pm 3\,\%$, flat at $4\times 8$) and the simpler collision adjoints (23 stored gates reproduce to $7\times 10^{-10}$).

# The per-cell interpolants tried

One least-squares polynomial per collision cell in scaled offsets $o = (\tilde{p} - \bar{p}_c)$ from the cell's node mean ($\tilde{p} = (x/\Delta x, v/v_s)$), relative ridge $10^{-8}$; `cell_fit.py`, `basis="cell:<poly>[:opt]"`.

| basis | what it is |
|---|---|
| `mean`, `lin` | constant (1 column); $1, o$ (6 in 2-D) |
| `quad` | the per-query fit's design: $1, o, v \otimes v, x_d^2$ (14) |
| `quadfull` / `cubicv` / `cubic` | every product (21) / + cubic in $v$ (24) / every monomial to order 3 (56) |
| `adapt`$a$ | **order guard**: highest order whose cumulative columns $\leq$ count/$a$; else a plane, else the mean |
| `loo` | per cell the order (0..max) of least leave-one-out residual $\sum w\,(r_i/(1-h_{ii}))^2$ |
| `pool`$r$ | fit the $(2r+1)^D$ block, tricube weight in $x$ (support $r+1$ cells), evaluate in the cell |
| `rb` | partner ghosts averaged over the VSS angle: 6 sigma points, exact for quadratics (any $\alpha$) |
| split (normals) | three fits per cell: leaving / approaching the surface ($v \cdot n_c \gtrless 0$) / all |
| `align`$r$, `both` | cells with $\lvert\sum n_i\rvert/\#n_i < r$ (complex): per-query fallback, or two-sided (`both`) |

The per-query fit's weights cannot be reproduced per cell (they are centred on the query); the per-cell fit is unweighted (or $x$-weighted when pooled).

# Correctness of the machinery

- every basis reproduces its own polynomials to $10^{-15}$; the fit is the numpy least-squares solution to $10^{-13}$; the slopes are the exact polynomial derivative ($4 \times 10^{-11}$);
- order guard: 60, 40, 25 nodes $\rightarrow$ quadratic, 14 $\rightarrow$ plane, 3 $\rightarrow$ mean; leave-one-out picks the mean on pure noise and the quadratic on a quadratic field;
- sigma points against $4 \times 10^5$ VSS scatterings: quadratics within 1.3 se at $\alpha = 1, 1.4$; cubics only at $\alpha = 1$ ($67$ se off at $1.4$, as derived);
- split fits in **1, 2 and 3-D** (collide-only box, $3^D$ cells): with every cell complex the hybrid backward *is* the per-query one-sided backward (relative difference $0$); fallback capacity = the measured load;
- an empty side (no node leaving the surface) gave NaN on a real geometry: now the identity, deferring to the unsplit fit (`test_cell_fit.py`, 12 tests).

**0-D relaxation** ($t_x$, 24 seeds, CRN FD): `cell:quad` $+0.0 \pm 3.0\,\%$ vs FD, $+0.3 \pm 0.7\,\%$ paired vs $k = 32$; backward 1.2 s vs $\approx 1100$ s (CPU). A `lin` fit gives $\approx 0$: $1, v$ are collision invariants, so a linear field has no collision term (the machinery's own check).

# Kn 0.5 squash, 12 seeds: the decisive comparison

$s = 1$, $f_{\mathrm{num}} \times 4$, $4 \times 8$, CRN FD $h = 0.2$ per seed: $0.01597 \pm 0.00026$.

| backward field | $\mathrm{d}D/\mathrm{d}s$ | vs FD | $-$ ($k$=64): sd (max) | s / seed, bwd |
|---|---|---|---|---|
| per-query $k = 32$ | $0.01541 \pm 0.00025$ | $-3.5 \pm 1.8\,\%$ | 2.5 (6.0) % | 670 / ≈550 s |
| per-query $k = 64$ | $0.01565 \pm 0.00030$ | $-2.0 \pm 2.0\,\%$ | — | 985 / 897 s |
| `cell:quad` | $0.01665 \pm 0.00107$ | $+4.3 \pm 7.9\,\%$ | **28 (95) %** | 98 / 13 s |
| `cell:quad:rb` | $0.01544 \pm 0.00057$ | $-3.3 \pm 4.0\,\%$ | 11 (34) % | 112 / 27 s |
| `cell:quad:adapt1.5` | $\mathbf{0.01566 \pm 0.00024}$ | $\mathbf{-1.9 \pm 2.3}\,\%$ | 5.3 (13) % | **95 / 13 s** |
| `cell:cubicv:adapt1.5` | $0.01611 \pm 0.00019$ | $+0.9 \pm 1.8\,\%$ | 5.5 (15) % | 103 / 20 s |
| `cell:quad:pool1` (3 seeds) | $0.01452 \pm 0.00030$ | $(-10.9 \pm 4.4\,\%)$ | 3.8 (8) % | 131 / 53 s |

- **Unguarded** `quad` **is unstable**: seed 11 $+97\,\%$, seed 10 $-26\,\%$ (the per-query fits: $+2$, $-8\,\%$). Sparse cells' underdetermined quadratics answer the partner ghosts with extrapolated noise and the recursion compounds it. `rb` (seed 10 $-40\,\%$) does not cure it.
- **The order guard does**: `adapt1.5` matches $k = 64$ ($+0.05 \pm 1.5\,\%$) and the FD at $\frac{1}{10}$ of the seed's time; its seed-to-seed SE equals the per-query fit's.

# Per seed: adjoint / FD $-1$, %

| seed | FD | $k = 32$ | $k = 64$ | quad | quad:rb | quad:adapt1.5 | cubicv:adapt1.5 |
|---|---|---|---|---|---|---|---|
| 0 | 0.01532 | $-9.5$ | $-5.8$ | $-2.8$ | $-0.7$ | $-5.0$ | $-3.2$ |
| 1 | 0.01744 | $-13.4$ | $-11.0$ | $-10.6$ | $-10.0$ | $-17.7$ | $-7.9$ |
| 2 | 0.01610 | $-7.7$ | $-4.4$ | $-3.5$ | $-3.3$ | $-5.4$ | $-1.8$ |
| 3 | 0.01629 | $+1.8$ | $+3.2$ | $+2.7$ | $+0.9$ | $+3.6$ | $+5.4$ |
| 4 | 0.01636 | $+0.7$ | $+3.2$ | $+3.7$ | $+3.1$ | $+1.5$ | $+4.9$ |
| 5 | 0.01547 | $-0.3$ | $+3.0$ | $+3.4$ | $+5.2$ | $+1.2$ | $+4.8$ |
| 6 | 0.01596 | $-5.1$ | $-7.1$ | $-7.0$ | $-6.3$ | $-7.8$ | $-1.7$ |
| 7 | 0.01598 | $+3.0$ | $+9.5$ | $+13.8$ | $+14.3$ | $+2.9$ | $-0.3$ |
| 8 | 0.01536 | $+5.2$ | $+4.7$ | $+4.6$ | $+5.0$ | $+3.9$ | $+6.2$ |
| 9 | 0.01686 | $-9.2$ | $-9.9$ | $-10.1$ | $-10.5$ | $-5.1$ | $-3.0$ |
| 10 | 0.01655 | $-8.0$ | $-8.3$ | $\mathbf{-25.5}$ | $\mathbf{-39.8}$ | $-5.5$ | $-5.7$ |
| 11 | 0.01393 | $+2.3$ | $+1.2$ | $\mathbf{+97.2}$ | $+6.2$ | $+14.2$ | $+16.5$ |

The per-seed spread is mostly the FD's own (CRN FD at $h = 0.2$, $\approx \pm 6\,\%$ per seed); the paired columns move together except where an unguarded fit breaks.

# Variance: what `rb` buys, and why value cross-validation misleads

**The backward's own noise** (0-D relaxation, 8 forwards × 8 partner keys, sd of the gradient):

| | sampled angle | `rb` (exact angle average) |
|---|---|---|
| $4 \times 8$ | 1.37 % | 0.91 % |
| $4 \times 2$ | 2.95 % | 1.85 % |

Variance $/2.2$ — real, but small against a forward's seed-to-seed noise ($\approx 3.5\,\%$), and no help against the sparse-cell instability.

**5-fold cross-validation of node values** (squash snapshot, $k = 64$ backward, 3 steps): the values are nearly pure noise — the cell mean predicts held-out nodes at the noise floor (mse $3.0 \times 10^{-12}$, variance $3.1 \times 10^{-12}$); every unguarded quadratic overfits to $3\times$ ($k = 32$: 9.7, $k = 64$: 8.9, per-cell `quad`: 9.3, all $\times 10^{-12}$), mostly in cells of $< 20$ nodes; guarded / pooled / LOO fits reach 2.2–3.5. **But this cannot rank fits for the backward**: the cell mean wins it and has *no collision term* ($1, v, |v|^2$ are invariants; only the rest of $\lambda$ enters $\mathcal{L}^*\lambda$). Only gradients against FD rank them.

# The stored collisional gates, re-run with the per-cell fits

Paired with the gates' own per-query $k = 32$ runs (same forwards; `scratch/cellfit/gates*.jsonl`).

| gate (seeds) | `cell:quad` vs $k = 32$ | `adapt1.5` | `rb` |
|---|---|---|---|
| 0-D relaxation $t_x$ (24) | $+0.3 \pm 0.7\,\%$ | — | $+0.3\,\%$ vs FD |
| Couette speed: energy, shear windows (4; 32) | $\leq 0.2 \pm 0.6\,\%$ | same | same |
| Couette speed, one-sided split (2; 32) | $+0.2 \pm 0.6$, $+0.2 \pm 0.2\,\%$ | same | same |
| Couette wall temperature (32) | $+0.3 \pm 1.3\,\%$ | same | $+0.5 \pm 1.3\,\%$ |
| Couette temperature, one-sided (2; 32) | $+4.5 \pm 7.0$, $+5.0 \pm 5.4\,\%$ | same | same |
| inlet $n$ / $T$, $N = 128$ (32) | $+0.7 \pm 0.3$ / $+2.0 \pm 1.1\,\%$ | $+0.8 \pm 0.6$ / — | $+0.3$ / $+1.5\,\%$ |
| relaxation `diameter` / `omega` (8) | $+1.2 \pm 0.3$ / $-0.3 \pm 0.6\,\%$ | same | — |

These cells hold $\leq\approx 40$ nodes (the order guard rarely engages); the FD noise of these gates ($\pm 6$ to $\pm 280\,\%$) is far larger than every difference. Each per-cell run took seconds where the per-query fit took up to 40 minutes on the CPU.

# Cut cells on real geometry: survey and the one-sided failure

**Geometry survey** (optc iterates, normal resultant $R$ per cell over its flow region): cells with $R < 0.9$ hold $\leq 6\,\%$ of the flow (circle: 0), $R < 0.8 \leq 5\,\%$; up to 24 cells contain opposed normals.

**optc S1-adam iterate 100, Kn 0.5** ($f_{\mathrm{num}} \times 4$; $g \cdot \mathbf{1}$ = derivative along the uniform level-set shift; FD $h = 0.05$):

| backward field | seed 1 | seed 2 |
|---|---|---|
| FD | $-0.0046$ | $-0.0055$ |
| per-query, one-sided $k = 32$ (the old option) | $+0.113$ ($\lambda$ rms $\times 150$) | — |
| per-query, two-sided $k = 32$ | $-0.0024$ | $+0.018$ |
| `cell:quad`, two-sided (unguarded) | $-0.0028$ | $-0.027$ |
| `cell:quad:adapt1.5`, two-sided | $\mathbf{-0.0049}$ | $\mathbf{-0.0044}$ |
| `cell:quad:adapt1.5`, split everywhere | $-0.0040$ | $+0.0033$ |
| `cell:quad:adapt1.5:align0.9:both` (shipped) | $\mathbf{-0.0035}$ | $\mathbf{-0.0038}$ |

The circle at the same settings: every fit sane. Falling back to the per-query one-sided fit in complex cells inherits its divergence; the shipped choice leaves those cells two-sided. Two seeds of one shape: evidence, not a gate — the cluster `cellgate` runs six shapes × 16 seeds.

# Selection

**Shipped** (`ensemble_backward.CELL_BASIS`, 2026-10-11): `cell:quad:adapt1.5:align0.9:both`

- one quadratic per cell, its columns capped at count/$1.5$ (else plane, else mean);
- with surface normals: split at $v \cdot n_c = 0$ (each side guarded), empty sides and cells with $R < 0.9$ two-sided;
- matches the 64-nearest fit ($+0.05 \pm 1.5\,\%$) and FD ($-1.9 \pm 2.3\,\%$); seed-to-seed SE unchanged; every stored gate; stable on the optc shapes where the per-query fits failed;
- **a squash seed** $985 \to 95$ **s** (backward $897 \to 13$ s): $\sim 10\times$ more optimisation iterates per GPU-hour.

**Rejected**: unguarded `quad`, `rb` (heavy tails); `pool1` (reads low, 3 seeds); the per-query fallback (unstable one-sided fit). **Kept as options**: `cubicv:adapt1.5` ($+3\,\%$ vs $k = 64$, $+0.9\,\%$ vs FD: not distinguishable at 12 seeds), `rb`, `loo`, `pool`, `own`, `quadratic_full`. The shape gradient's *slope* fits (pooled, one-sided, $k = 96$) are unchanged.

**Open**: `cellgate` on the cluster at production density ($f_{\mathrm{num}} \times 1$): 16 + 8 squash seeds (all fits paired with FD), six optc shapes at Kn 0.5 and free-molecular, two FD directions.

# Validation status of the current estimator

| case | result | regime |
|---|---|---|
| parameter gradients (inlet, Couette walls, VSS kernel), virtual $4 \times 8$ | $\lvert z\rvert \le 1.7$; $\le 1\,\%$ bias | collisional |
| Rankine–Hugoniot $\mathrm{d}(\rho_2/\rho_1)/\mathrm{d}M_1$ | $z = 0.3$ (coarse) | collisional |
| level-set circle radius; sphere radius | $z = -1.4$; $z = -1.71$ | free-mol. |
| $\cos 2\theta$ coarse, state-based fits | paired $+0.007 \pm 0.071$ | free-mol. |
| diffuse plate (silhouette + hybrid normal) | $z = +0.19$ ($\lvert z\rvert \le 1.6$ at 5 steps) | free-mol. |
| squash specular, $s = 0.35/0.6/1$ | $-1.7/-2.5/-1.5\,\%$ ($\pm 1.5$); $\approx -1\,\%$ is horizon | free-mol. |
| squash $s = 1$, diffuse | $-1.4 \pm 3.3\,\%$ | free-mol. |
| cut-volume machinery | $z = -1.2$ (all cells), $+0.8$ (cut cells) | Kn 0.5 |
| squash $s = 1$, Kn $= 0.5$ (2026-10-03, $f_{\mathrm{num}} \times 2$, $k = 32$) | $-6.5$ to $-8.3 \pm 1.8\,\%$ | superseded |
| squash $s = 1$, Kn $= 0.5$, 12 seeds: per-cell `adapt1.5` / $k = 64$ | $-1.9 \pm 2.3$ / $-2.0 \pm 2.0\,\%$ | Kn 0.5 |

Free-molecular shape gradients, specular and diffuse, agree with FD to $\sim$2 %, and most of that is the captured horizon. The collisional gradient was 7–8 % low (8 seeds, $f_{\mathrm{num}} \times 2$); over 12 seeds at $f_{\mathrm{num}} \times 4$ the per-query $k = 32$ reads $-3.5 \pm 1.8\,\%$, $k = 64$ and the shipped per-cell fit $-2\,\%$ ($\pm 2$). Parameter gradients on collisional flows are validated to $\le 1\,\%$, with either field fit.

# Open items

1. **Kn 0.5 gap** (now $\approx 2 \pm 2\,\%$): cluster `cellgate` (squash at $f_{\mathrm{num}} \times 1$, every fit paired with FD, 16 + 8 seeds; six optc shapes) and `biasgate` (`kn_fd*` 64-seed FD, horizons; its 2026-10-04 jobs all died on `import jax_dsmc`: fixed in `env.sh`, preflight added).
2. **Free-molecular remainder** ($\sim$1 % after 600 captured steps): `fm_hor_s*`.
3. **Re-runs**: free-molecular squash and `optc` can be re-run now (the old runs used `along_pool`, $-9\,\%$ at $s = 0.35$, and `optc` the future class); squashkn after item 1.
4. **optc** bodies need a body-to-face clearance constraint before their drag reductions are quoted: the `optc2` and `optckn` re-runs use a 3× longer box centred on the body (`circle_long`, 15 m each side; warm-up 2800 steps) and keep the body 1 m from every face (`optimize --keepout`).
5. Translations of flat walls remain unsupported as controls.

Summaries: `scratch/bias/summarise.py`, `horizon_summary.py`; cluster: `examples/mesh/cluster/README.md` (`biasgate`).